



HWA CHONG INSTITUTION (COLLEGE)
JC2 H1 PRELIMINARY EXAMINATIONS 2009
BIOLOGY 8875 – MARK SCHEME

Paper 1 (CORE and APPLICATIONS)
MCQ

1. A	6. B	11. B	16. B	21. D	26. B
2. B	7. B	12. A	17. D	22. A	27. A
3. C	8. C	13. D	18. B	23. B	28. C
4. D	9. B	14. B	19. A	24. D	29. D
5. B	10. D	15. C	20. B	25. A	30. A

Paper 2 (CORE)
Section A: Structured

QUESTION 1

(a)(i) Define the term glycoprotein.

[1]

- Protein / polypeptide;
- with oligosaccharides /carbohydrates;
- covalently attached;

(1m max)

(ii) Suggest a role of PrP in the membranes of nerve cells.

[1]

- Cell signaling / cell membrane receptors receiving signals from ligands;; OR Cell adhesion;;

(1m max)

(b)(i) With reference to **Fig. 1.1 & Fig. 1.2**, relate how the differences in the structure of PrP and PrP^{SC} bring about the differing effects of enzyme action.

[2]

PrP^{SC} has

- higher proportions of β -sheets than PrP;
- i.e. 4 in PrP^{SC} as compared to 1 in PrP;

OR

- lower proportion of α -helices than PrP;
- i.e. 2 in PrP^{SC} as compared to 3 in PrP;

(1m max)

- There are differences in the folding / bending / twisting; of the PrP and PrP^{SC} proteins
- Which give rise to different 3-D conformations;;
- Protein is no longer complementary to the active site of enzyme;
- Protein unable to form enzyme-substrate complex/bind to the active site of enzyme;

(1m max)

(c) With reference to **Fig 1.3**,

(i) identify the evidence that suggests the experimenters tested only one sample in each of the controls.

[1]

- Absence of error bars;
- No variation observed;
- i.e. % PrP^{SC} after 48 hrs remained at 100%; for both mouse and human
- at 0:1 ratio;

(1m max)

(ii) suggest **two** conclusions about the effect of β -breaker on human and mouse PrP^{SC}.

[2]

Any two of the following:

β -breaker decreases % PrP^{SC}



- As ratio of β -breaker to PrP^{SC} increases, % PrP^{SC} decreases in mice and human;
 - from 100% at ratio 0:1 to 15% at ratio 1000:1 for mice;
 - and from 100% at ratio 0:1 to 37.5% at ratio 100:1 for human;
- (1m max)

Comparison between mice and human.

- % PrP^{SC} present after 48 hours is more in human than mice at all ratios;
 - There is a bigger % PrP^{SC} decrease in mice than humans for each ratio;
 - thus, the β -breaker works better/greater effect in mice;
- (1m max)

Mice

- In mice there is less % PrP^{SC} decrease as β -breaker concentration increases;
 - There is a drop of % PrP^{SC} from 52.5% at ratio 10:1 to 19.5% at ratio 100:1 but a drop from 19.5% at ratio 100:1 to 15% at ratio 1000:1 OR There is a drop from 100% at ratio 0:1 to 50% at ratio 1:1 but a drop from 19.5% at ratio 100:1 to 15% at ratio 1000:1;
 - Which indicates that β -breaker gets less effective at higher concentrations;
 - β -breaker is less effective for mice at 10:1 ratio as % PrP^{SC} present showed an increase from 50% to 52.5%;
- (1m max)

Human

- Higher concentration of β -breaker may not be more effective with decreasing % PrP^{SC} ;
 - Best effect in humans is 100:1 when the % PrP^{SC} decrease to 37.5%;
 - Larger variation in digestion of human PrP^{SC} ;
 - There is a drop of % PrP^{SC} from 100% at ratio 1:1 to 37.5% at ratio 100:1 but a increase from 37.5% at ratio 100:1 to 50% at ratio 1000:1;
- (1m max)

(d) With reference to **Table 1.4**,

(i) Explain how two conclusions can be derived from the trends observed.

[3]

Concentration of PrP^{SC}

- The lower the concentration of PrP^{SC} the greater the delay in onset of symptoms;
 - As concentration of PrP^{SC} increases from 1×10^{-2} to 1×10^{-4} , the time to onset of symptoms increases from 129 days to 173 days;
 - PrP^{SC} is an infectious agent;
- (1.5m max)

Presence of β -breakers.

- The presence of β -breakers increases the delay in onset of symptoms;
 - for all concentrations of PrP^{SC} from 143 days to 185 days;
 - The β -breaker is effective in delaying the onset of the symptoms;
 - Once infected by PrP^{SC} , there is still no cure;
- (1.5m max)

[Total: 10 marks]

QUESTION 2

(a)(i) State a possible identity for this gene regulatory protein.

[1]

- (General) transcription factor;; OR Activator;;

(ii) Describe how this gene regulatory protein identified in **(a)(i)** could result in higher transcriptional efficiencies.

[2]

- Binding of this (general) transcription factor to the core promoter / promoter-proximal elements;
- OR
- Binding of activator to enhancer;
 - recruits other DNA-binding proteins / transcription factors / activators / co-activators;
 - to form a stable transcription initiation complex; (resulting in an increase in transcription levels)



(b)(i) With reference to **Fig. 2.1** and **Fig. 2.2**, compare the effects of the different deletions between the purified components and the nuclear extract. [2]

- Deletion up to 11 bp upstream of transcription start site / -11 deletion template inactivates transcription in both systems;
- Deletions up to 61 bp upstream of transcription start site / -61, -81, -127 deletion templates have no effect on transcription level / results in the same transcription level as compared to the undeleted template in both systems;
- Deletion up to 50 bp upstream of transcription start site / -50 deletion template results in a lower transcription level in the nuclear extract; which is comparable to the transcription level for deletions up to 50 bp upstream of transcription start site / -50 deletion template in the purified system;

(ii) Account for the absence of a band for the -11 deletion template in both the purified system and the nuclear extract as shown in **Fig. 2.2**. [2]

- Deletions up to 11 bp upstream of transcription start site / -11 deletion template would imply that the TATA box (which resides ~30 bp upstream of transcription start site) is deleted;
- meaning that general transcription factor cannot bind; resulting in no recruitment of RNA polymerase to the core promoter; and thus no transcription / no mRNA transcripts formed; (resulting in no bands formed for both systems)

(c)(i) State the type of gel electrophoresis used. [1]

- Sodium dodecyl sulphate – polyacrylamide gel electrophoresis / SDS-PAGE;;

(ii) Explain how the discrete bands shown in **Fig. 2.3** are formed. [2]

- Protein components of SAGA complex are separated by / on the basis of size;
- where gel matrix acts as a molecular sieve; for migration of unfolded protein (tagged with negatively-charged SDS) towards the anode / positive electrode upon application of an electric field;
- Larger proteins would be more impeded / move slower than smaller proteins;
- thus when stained with Commassie Blue; forming discrete bands on the gel (2m max)

(iii) The estimated molecular weight of protein X using gel electrophoresis shown in **Fig. 2.3** is 58 kd. However, its actual molecular weight is only 46 kd. Suggest a possible reason for this discrepancy. [1]

- Protein X could contain a large number of inherent positive charges; AND
- which would partially offset the negative charges added by SDS; OR
- and would carry less overall negative charges than would an average SDS-coated protein of its size; OR
- thus run anomalously slowly on SDS gel;

[Total: 11 marks]

QUESTION 3

(a)(i) Deduce the inheritance pattern of the disease. [1]

- Autosomal; recessive;

(ii) Explain how **Fig. 3.1** provides evidence to justify your answer in part **(a)(i)**. [2]

- Two unaffected parents II-3 and II-4 produced affected offspring III-1; who inherited a copy of the disease allele from each parent; OR
- Each of the two parents II-3 and II-4 is heterozygous / a carrier; and transmits a copy of the disease allele to affected son III-1; indicating that the disease is recessive.
- Then, if the disease is sex-linked, affected mother I-2 will pass the disease allele to all her sons; but both her sons II-1 and II-2 are unaffected;

(b) Individual **V-3** married individual **V-4** and the couple plans to have a child. Calculate the probability that their first child will be affected with the disease. [1]

- 1/16;;



(c) Explain why affected individuals in Generation I can produce normal children. [2]

- Affected individuals are homozygous recessive; and married normal individuals who are either homozygous dominant; or heterozygous carriers;
- thus at least one copy of the normal dominant allele is transmitted to their children; who will thus be phenotypically normal as the dominant normal allele from the unaffected parent masks the effect of the recessive disease allele (from the affected parent);

(d) Using the symbols **A** and **a** to represent the alleles for skin pigmentation, and **B** and **b** to represent the alleles for colour vision, draw in the space below a genetic diagram of the cross between individuals II-3 and II-4. [4]

Let: A be the dominant allele for normal pigmentation
a be the recessive allele for lack of pigment ;
B be the dominant allele for normal vision
b be the recessive allele for colour-blindness ;

Parental Phenotypes Normal pigmentation, normal vision (female) x Normal pigmentation, colour-blind (male)

Parental Genotypes AaX^BX^b AaX^bY ;

Gametes AX^B AX^b aX^B aX^b AX^b AY aX^b aY ;

Random fertilisation

		Female gametes			
		AX^B	AX^b	aX^B	aX^b
Male gametes	AX^b	$AA X^B X^b$	$AA X^b X^b$	$Aa X^B X^b$	$Aa X^b X^b$
	AY	$AA X^B Y$	$AA X^b Y$	$Aa X^B Y$	$Aa X^b Y$
	aX^b	$Aa X^B X^b$	$Aa X^b X^b$	$aa X^B X^b$	$aa X^b X^b$
	aY	$Aa X^B Y$	$Aa X^b Y$	$aa X^B Y$	$aa X^b Y$

∴

Offspring genotypic ratio $3A_X^BX^b:3A_X^bX^b:1aaX^BX^b:1aaX^bX^b:3A_X^BY:3A_X^bY:1aaX^BY:1aaX^bY$;

Offspring phenotypic ratio 3 normal pigmentation, normal vision female : 3 normal pigmentation, colour-blind female : 1 lack of pigmentation, normal vision female : 1 lack of pigmentation, colour-blind female : 3 normal pigmentation, normal vision male : 3 normal pigmentation, colour-blind male : 1 lack of pigmentation, normal vision male : 1 lack of pigmentation, colour-blind male ;

[Total: 10 marks]

QUESTION 4

(a) With reference to Fig. 4.1, compare

(i) the mean index of tolerance to copper of adult plants to that of plants grown from collected seed; [1]

- Mean index of tolerance to copper is higher in adult plants than seedlings in both samples;



- Uncontaminated soil, adult mean index is 35 as compared to 29 in seedlings / higher by 6; OR contaminated soil, adult mean index is 69 compared to 45 / higher by 25;.

(ii) the range in tolerance to copper of the adult plants with plants grown from collected seed. [1]

- Range in tolerance to copper is smaller / narrower in adult plants than seedlings in both samples;
- Uncontaminated soil, range of index of adult plants is 50 as compared to 80 in seedlings / smaller by 30; OR contaminated soil, adult mean index is 30 compared to 70 / narrower by 40;.

(b) Account for the variation in the plants grown from seed. [2]

Variation could be due to:

- polygenic control; thus continuous variation;
- genetic variation; as a result of due to crossing over / random assortment / random fertilization; during meiosis;
- random fertilization of gametes;

(1.5m max)

AND

These will result in a larger variation in plants grown from seeds compared to the parent plants;

(c) Explain why the same variation is not found among the plants growing on the contaminated soil. [3]

- Plants with copper tolerance are better adapted; and have a selective advantage; in contaminated soils and are thus selected for;
- These plants have a higher chance of survival and reproduction; pass on the beneficial alleles to their offspring / like producing like; thus these plants produce seedlings with high tolerance;
- The allelic frequency of the copper-tolerance allele will increase; in the plant population growing in contaminated soil while that of copper-intolerance allele will decrease;

(d) Suggest how the results of this study could be used in the reclaiming of contaminated land. [2]

- Copper tolerant plants could be selected; to be grown on contaminated soil
- in order to reclaim land for agricultural use;;

[Total: 9 marks]

Paper 2 (CORE and APPLICATIONS)

Section B: Free Response

QUESTION 5

(a) Describe the synthesis of ATP in photosynthesis and respiration. [12]

Photosynthesis

1. Occurs at the thylakoid membranes;
2. occurs in the light dependent stage/photophosphorylation in photosynthesis ;
3. Photosystems harvest light energy and channels it to the primary pigment to generate high energy electrons ;
4. which will passed down the electron transport chain to NADP if electrons originate from PSI or PSI if electrons originate from PSII ;
5. Energy released will be coupled to the active transport/pumping of the protons ;
6. from the stroma into the thylakoid lumen ;
7. The accumulation of protons in the thylakoid lumen ;
8. generates a proton / electrochemical gradient ;
9. ATP synthases / stalked particles allow for the flow of protons into the stroma whereby the energy released is coupled to;
10. the phosphorylation of ADP to form ATP / chemiosmosis;

Respiration

11. Occurs in the mitochondrial matrix and inner mitochondrial membranes;
12. In glycolysis, 1 molecule of glucose is oxidized to 2 molecules of pyruvate with net production of 2 ATP ;
13. via substrate level phosphorylation ;
14. With every turn of the Krebs cycle, 1 ATP molecule is produced;
15. by substrate level phosphorylation ;



16. Electrons are stripped from respiratory intermediates of glycolysis and Krebs cycle;
17. are brought to the electron transport chain by reduced co-enzymes NADH and FADH₂;
18. Electrons move from low- to high-affinity electron carriers to final electron acceptor oxygen ;
19. Energy released is coupled to the active transport/pumping of the protons ;
20. into the inter-membrane space ;
21. The accumulation of protons in the inter-membrane space ;
22. generates a proton / electrochemical gradient ;
23. ATP synthases / stalked particles allow for the exergonic flow of protons into the mitochondria matrix whereby the energy will be coupled to;
24. the phosphorylation of ADP to form ATP/ chemiosmosis ;
25. 34 ATPs generated per glucose molecule in the electron transport chain;

(b) Discuss the significance of co-enzymes involved in photosynthesis and respiration.

[8]

Photosynthesis

1. NADP is the co-enzyme in photosynthesis and is the final electron acceptor of the non-cyclic photophosphorylation ;
2. which is essential for the continuous movement of electrons down the electron transport chain ;
3. to allow for the synthesis of ATP by photophosphorylation;
4. ATP is a source of energy ;
5. and NADPH is the source of reducing power to drive the Calvin cycle ;
6. to allow for carbon dioxide fixation ;
7. to generate glyceraldehyde-3-phosphate/PGAL/TP ;
8. which is the starting substrate for the formation of the various biomolecules in plants eg: carbohydrate/glucose, proteins, fats, nucleic acid and chlorophyll ;

Respiration

9. In glycolysis, glyceraldehyde-3-phosphate is oxidised to 1,3-bisphosphoglycerate ;
10. while the electron acceptor NAD is reduced to NADH ;
11. NADH is also generated during the oxidative decarboxylation of pyruvate to acetyl-CoA ;
12. which enters the Krebs cycle ;
13. In the Krebs cycle, high-energy electrons are stripped from respiratory intermediates;
14. and donated to NAD and FAD to generate NADH and FADH₂ ;
15. Mobile carrier NADH and membrane-bound FADH₂ channels the electrons down the ETC ;
16. allowing for ATP synthesis by oxidative phosphorylation / chemiosmosis ;
17. ATP produced is essential to power the various cellular processes in the body e.g. muscle contraction and active transport etc ;

[Total: 20 marks]

QUESTION 6

(a) Describe the natural functions of restriction enzymes and explain how they can be used in the process of gene cloning. [10]

1. REs each binds to a double-stranded DNA molecule;
2. at a specific DNA sequence / restriction site;
3. and makes a double-stranded cut at or near that sequence;

Natural functions of REs

4. REs are naturally found in and isolated from bacteria;
5. Natural function of REs is to provide a protective mechanism;
6. to help bacteria resist attack from bacteriophages;
7. and to help in the removal of viral sequences;
8. REs work by recognizing specific sequences in the phage DNA and cleaving those sequences;
9. thereby degrading the incoming viral DNA;
10. thus preventing / restricting the bacteriophage from succeeding in causing a full-blown bacterial infection;

How REs can be used in gene cloning

11. A chosen RE is used to cleave the gene donor / chromosome / cDNA library;
12. to isolate the gene of interest;
13. The same RE is used to cleave the cloning vector / plasmid;
14. making the plasmid DNA linear;
15. to facilitate subsequent insertion of the excised gene of interest into plasmid;



16. Usually, the chosen RE will cleave the DNA in a staggered manner;
 17. resulting in short single-stranded overhangs / sticky / cohesive ends at each end of the molecule;
 18. sticky-ended fragments of gene of interest are incubated with complementary sticky-ended linearized vector fragments under conditions that favour annealing;
 19. so as to facilitate the formation of a recombinant plasmid by DNA ligase;
 20. If the chosen RE make a cut in the middle of the recognition sequence;
 21. resulting in blunt / flush ends;
 22. adaptors will have to be ligated by DNA ligase to the ends of gene of interest and plasmid fragments;
 23. to generate sticky-ended fragments;
- (b) Outline how **two** named proteins can be produced by the genetic engineering technique using different host organisms. [10]

Named example 1

1. Synthesis of human growth human (hGH) / somatotrophin;
 2. in bacteria, *Escherichia coli*;
 3. Mature mRNA is obtained from the pituitary gland that produces hGH;
 4. and reverse transcribed to prepare a pituitary cDNA library;
 5. which is subsequently digested with the restriction enzyme *HaeIII* into two fragments;
 6. The longer segment is retained for use in construction of the recombinant plasmid;
 7. while the smaller segment is replaced by an artificial DNA molecule;
 8. that contains the start of the hGH gene and provides the correct signals for translation in *E. coli*;
 9. The modified gene is ligated into a bacterial plasmid carrying the *lac* promoter;
 10. recombinant plasmids are introduced into *E. coli* cells either via heat shock or electroporation;
 11. Selection and identification of *E. coli* clones with recombinant plasmids carrying the gene of interest is carried out;
 12. via the use of selection marker genes (antibiotic resistance / and nutritional requirement);
 13. and nucleic acid hybridization;
 14. Identified *E. coli* clones with recombinant plasmids carrying the gene of interest were isolated and grown in liquid culture;
 15. Lactose is added to induce the production of hGH;
- (5m max)

Named Example 2

17. Synthesis of human anti-thrombin III (hATIII);
18. in eukaryotic cell lines;
19. Mature mRNA is obtained from liver cells that produce hATIII;
20. and reverse transcribed to form the hATIII cDNA;
21. which is ligated to a DNA construct containing a promoter recognized by the eukaryotic host cell;
22. The gene construct is subsequently inserted into a eukaryotic expression vector / eukaryotic virus;
23. which is used to transfect eukaryotic host cell lines;
24. The hATIII cDNA is then inserted into the eukaryotic host cell genome;
25. by homologous recombination;
26. Expression of hATIII cDNA takes place under the control of the eukaryotic promoter;
27. hATIII produced by eukaryotic cell lines is isolated for further purification;

[Total: 20 marks]